

Vinay Kumar Mandadi

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EDUCATION

Lovely Professional University

B.Tech in Computer Science and Engineering

Punjab, India

Expected 2026

- Relevant coursework: Advanced Web Development, JavaScript frameworks, and UI/UX principles; supported by a specialisation in Google Cloud and DevOps covering CI/CD, Infrastructure-as-Code, and Site Reliability Engineering.
- **GRE General Test:** 324 (Quantitative: 168, Verbal: 156)

PROFESSIONAL CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 — Certified Multicloud Architect Professional
- Oracle Cloud Infrastructure 2025 — Certified DevOps Professional
- Oracle Cloud Infrastructure 2025 — Certified AI Foundations Associate
- Oracle Cloud Infrastructure 2025 — Certified Data Platform Foundations Associate

TECHNICAL SKILLS

- **Frontend:** TypeScript (primary), JavaScript (ES6+), React (Hooks, Context API, state management, data fetching), HTML5, CSS3
- **Backend & Tooling:** Node.js, Express.js, MongoDB, Flask, Firebase, Google App Engine, Git, Docker, CI/CD pipelines, Unix/Linux
- **Cloud & DevOps:** AWS, GCP (Cloud SQL, Cloud Vision API, App Engine), OCI, Infrastructure-as-Code, Terraform
- **Languages:** English (professional proficiency) · Japanese (elementary; Basic Japanese I & II, Elementary Japanese III) · Telugu(native), Hindi

FRONT-END & FULL-STACK PROJECTS

StockAI — Stock Market Visualization Platform

 TypeScript · React · Firebase Hosting stockai338211.web.app

- TypeScript chosen from day one to enforce strict data-shape contracts between API responses and UI state — catching mismatches at compile time rather than runtime.
- Implemented React Context for global state management; data fetching with async/await, loading states, error boundaries, and graceful degradation for failed requests.
- Deployed to Firebase Hosting via automated CI pipeline — every merged commit reaches production without manual steps.

Carpooling and RideShare Application

 TypeScript · HTML · MongoDB · Node.js · Express Full-stack

- Built full-stack in TypeScript — HTML/CSS frontend and Node.js/Express backend — with client-side form validation on ride creation, booking, and review flows before any server round-trip.
- REST API contracts defined before UI components were written, keeping state management predictable and integration friction low across all client-rendered flows.
- Session-based authentication end-to-end: token refresh, redirect-on-expiry, and consistent error states across all protected routes.

Compass AI — Travel Itinerary Planner

 JavaScript · Firebase Hosting · Firebase Realtime DB Web app

- Static travel planning UI with Firebase Realtime Database for live itinerary sync across browser tabs without page reload.
- Loading skeleton components maintain perceived responsiveness during database reads; UI degrades gracefully when data is unavailable.

Secure Aadhaar Voting Platform

 GCP · App Engine · Cloud SQL · Cloud Vision API · Flask Group project

- Contributed to cloud infrastructure (App Engine auto-scaling, Cloud SQL, Cloud Storage, IAM least-privilege) and the Flask face-verification API calling Cloud Vision API for similarity scoring.
- HTTPS throughout; secrets in environment variables, not source code. PWA-style caching added for low-connectivity resilience.

RESEARCH & PUBLICATIONS

- **Deepfake Detection** (Co-author, accepted 8BIC 2026) — Dual-branch EfficientNetV2-M + DCT-CNN; 97–99% AUC-ROC and >93% accuracy on FaceForensics++ C23, >89% accuracy on Celeb-DF v2. GradCAM explainability overlays and full-stack Flask web app with bcrypt-authenticated upload pipeline.
- **AIRIMF: AI-Driven Risk Identification and Mitigation Framework** (Sole author, submitted to ICMACC 2026) — Leakage-free Random Forest pipeline predicting high-risk GitHub tickets at creation time using only submission-time features (NLP sentiment + structural metadata). 76.21% CV accuracy on VS Code (Class 2 recall 82%);

transfers to React at 84.70% without retraining. Cross-project calibration on Angular: Platt scaling cut ECE by 5× (0.2521 → 0.0501) and recovered 17.6 pp accuracy. SHAP confirms description length as the dominant risk signal. McNemar's test vs. SVM: $\chi^2 = 30.25$, $p < 0.0001$.

- **Explainable and Trustworthy Network Intrusion Detection: From Multi-Class SHAP Analysis to Confidence-Aware Deployment** (Sole author, submitted to PIMRC 2026 WS5) — Leakage-free multi-class IDS on CIC-IDS2017 (98.83% accuracy, PR-AUC 1.000 for Web Attack). Exposed catastrophic domain shift to UNSW-NB15 (31.9% accuracy, 0% attack recall); showed confidence-aware filtering fails under shift; Platt scaling recovered 68.1% OOD accuracy. Deployed as FastAPI endpoint and Streamlit dashboard with configurable confidence alerting.

ACHIEVEMENTS & COMMUNITY

- **GDG Solution Challenge:** National Finalist — Top 105 teams across India.
- **Google Cloud Skills Boost:** Gold Tier — self-directed learning running parallel to coursework, not replacing it.
- **Japanese:** Completed university coursework in Basic Japanese I, Basic Japanese II, and Elementary Japanese III; actively maintaining proficiency toward JLPT N4.